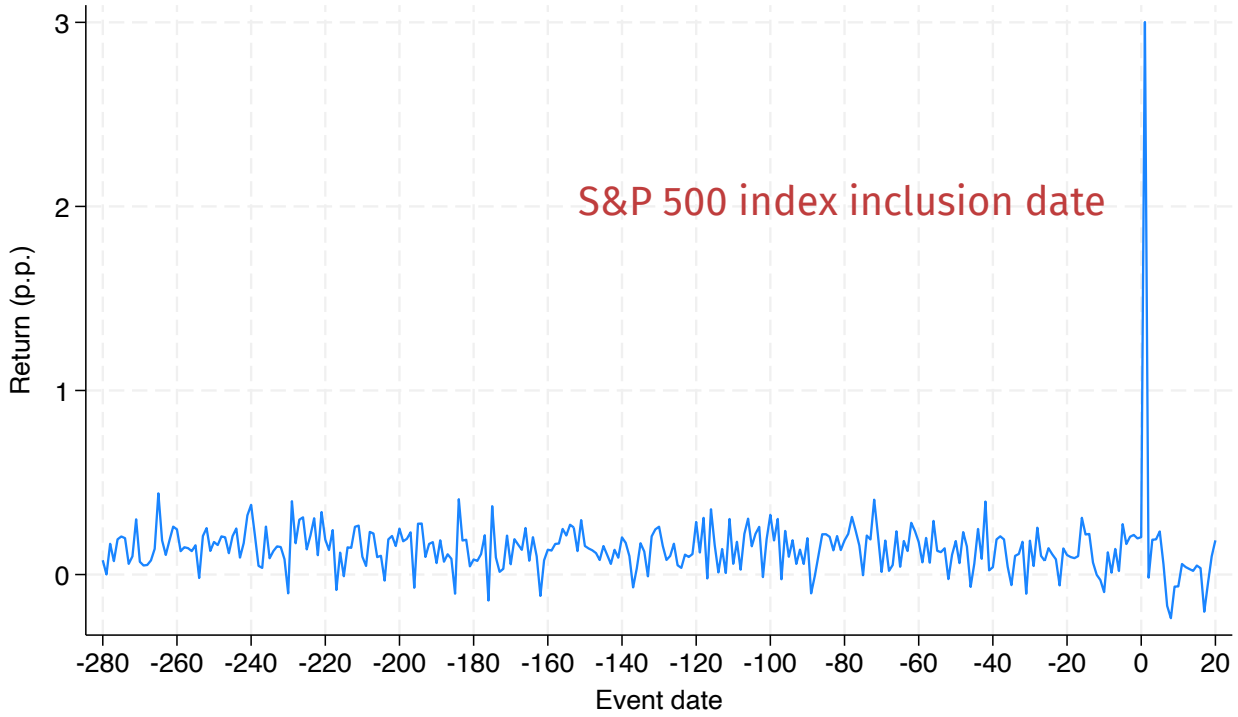
Causal Inference in Financial Event Studies

Paul Goldsmith-Pinkham (Yale SOM & NBER)
Tianshu Lyu (Yale SOM)

October 2026 https://paulgp.com

Tying together two literatures, and extending an old debate

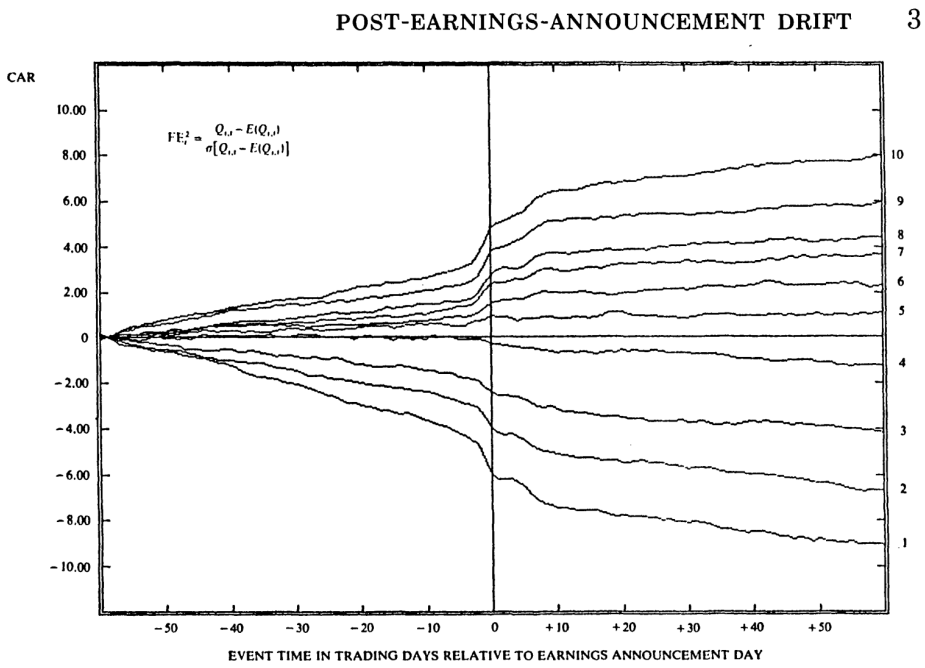
- Finance literature studying the impact of events on asset prices
- Econometrics literature estimating causal effects



Historically, financial event studies are an important tool

What types of financial events?
Examples...

- Index Inclusion
- Earnings Announcements
- Mergers and acquisitions
- IPO, SEO, Shares repurchased
- CEO/CFO Changes
- Patent Issuance
- FOMC Announcements
- Labor Issues
- Political events



Historically, financial event studies are an important tool

What types of financial events? Examples...

- Index Inclusion
- Earnings Announcements
- Mergers and acquisitions
- IPO, SEO, Shares repurchased
- CEO/CFO Changes
- Patent Issuance
- FOMC Announcements
- Labor Issues
- Political events



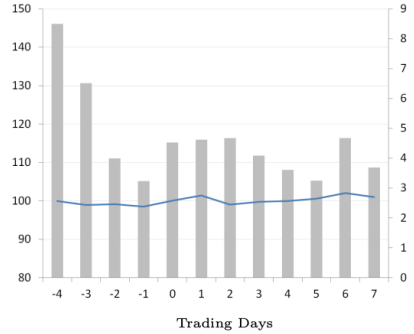
(a) Patent 4,946,778 granted to Genex on Aug, 7 1990, "Single Polypeptide Chain Binding Molecules."



(b) Patent 5,585,089 granted to Protein Design on Dec 17, 1996, "Humanized Immunoglobulins."

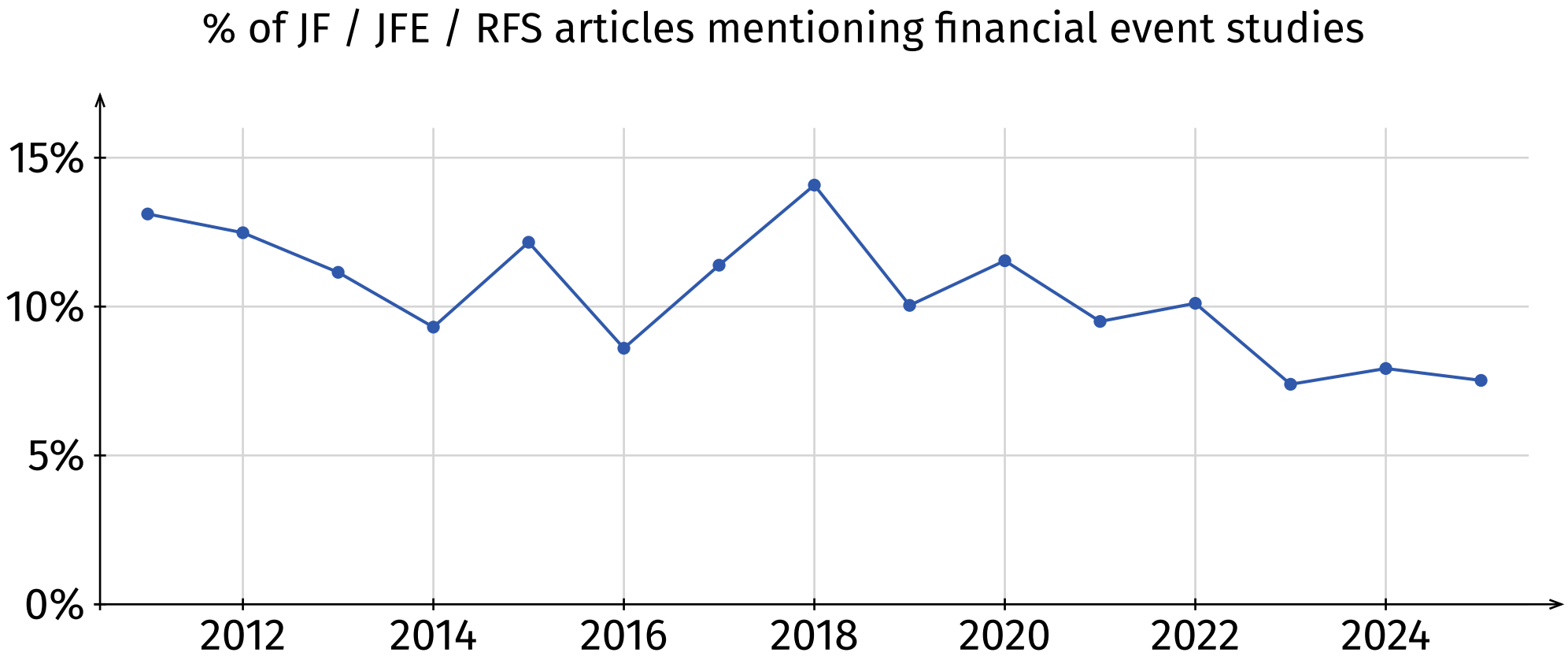


(c) Patent 6,317,722 granted to Amazon.com on Nov 13, 2001, "Use Of Electronic Shopping Carts To Generate Personal Recommendations."



(d) Patent 6,329,919 granted to IBM on Dec 11, 2001, "System and Method For Providing Reservations For Restroom Use."

Financial event studies continue to be used!



Source: https://paulgp.com/econlit-pipeline/search.html — search of JF, JFE, RFS articles mentioning “cumulative abnormal return[s]” or “announcement return[s]” or “buy-and-hold return[s]”.

Contribution of this paper

- Reframe event studies as a causal inference problem
  - What is the counterfactual return?
- Characterize when abnormal return estimators are biased
  - Short-run, many events: usually fine
  - Short-run, one event: depends on volatility
  - Long-run: almost always biased
- Propose alternative estimators
  - Synthetic control, PCA regression (GSynth)

- Four applications:
  1. Geithner / political connections
  2. S&P 500 Index Inclusion
  3. Mergers and acquisitions
  4. Close merger contests
- Revisiting established findings, we show some may reflect model misspecification rather than true treatment effects

Roadmap

1. The problem: abnormal returns and misspecification
2. Estimators and when bias matters
3. Four applications:
   - Geithner: single event, extreme volatility
   - S&P 500 Index Inclusion: many events, random timing
   - M&A: many events, long-run bias
   - Close contests: quasi-experimental benchmark
4. What should researchers do?

What is the effect of an event on stock returns?

- Unit of analysis: path of stock returns $R_i = \{R_{i1}, ..., R_{iT}\}$
  - $D_i \in \{0, 1\}$: event happens at t_0 to $n_0 < n$ firms
- Two potential versions of R_{it}:
  - $R_{it}(1)$: firm experienced the event
  - $R_{it}(0)$: firm without the event
- Researchers interested in the *causal effect* of the event:

$$\tau_{it} = R_{it}(1) - R_{it}(0)$$

- But, fundamental problem of causal inference:

$$R_{it} = R_{it}(1)D_i + R_{it}(0)(1 - D_i)$$

Placing a model on the structure of counterfactual returns

Textbook approach approximates with *abnormal returns* (Campbell, Lo, MacKinlay (1997))

$$AR_{it} = R_{it} - \underbrace{\mathbb{E}(R_{it} \mid X_t)}_{\text{Normal Returns given } X_t}$$

- $\mathbb{E}(R_{it} \mid X_t)$ reflects many models of expected returns (MacKinlay (1997))
  - Market Model, CAPM, Fama-French

$$R_{i,t} = \alpha_i + \underbrace{\beta_i^o}_{\text{loading}} + \underbrace{F_t^o}_{\text{observred factor}} + \varepsilon_{i,t}$$

- Estimate α_i, β_i^o on pre-event data
- Model captures every systematic factor $\rightarrow$ no issues

Key insight: asset-pricing models $\neq$ counterfactual prediction

Split returns into observed/priced factors and unobserved/unpriced factors:

$$R_{it} = \alpha_i + \beta_i^o F_t^o + \beta_i^u F_t^u + \varepsilon_{it}, \quad \mathbb{E}(F_t^u) = 0$$

Abnormal returns target expected returns:

$$\mathbb{E}(R_{it}) = \alpha_i + \beta_i^o \mathbb{E}(F_t^o)$$

- F_t^u drops out — it earns no premium
- Many random events → F_t^o averages out too, so the model "works"

Counterfactual needs the contemporaneous return:

$$E(R_{it}(0) \mid \text{event t}) = \alpha_i + \beta_i^o F_t^o + \beta_i^u F_t^u$$

- Match realized common shocks
- Includes unpriced and omitted factors

Model can be right for expected returns but *wrong* for counterfactual

Misspecification in the abnormal return estimator

What happens when the factor F_t^u is omitted?

- Single firm's abnormal return:

$$\begin{aligned}\widehat{AR}_{it} &= R_{it} - \hat{\alpha}_i - \hat{\beta}_i^o F_t^o \\ &= [\alpha_i - \hat{\alpha}_i] + \left[\beta_i^o F_t^o - \hat{\beta}_i^o F_t^o\right] + \beta_i^u F_t^u + \tau_t + \varepsilon_{i,t}\end{aligned}$$

Misspecification in the abnormal return estimator

What happens when the factor F_t^u is omitted?

- Single firm's abnormal return:

$$\begin{aligned}\widehat{AR}_{it} &= R_{it} - \hat{\alpha}_i - \hat{\beta}_i^o F_t^o \\ &= [\alpha_i - \hat{\alpha}_i] + \left[\beta_i^o F_t^o - \hat{\beta}_i^o F_t^o\right] + \beta_i^u F_t^u + \tau_t + \varepsilon_{i,t}\end{aligned}$$

- Average across all treated firms:

$$\overline{AR}_t \approx \tau_t + \underbrace{\left(\overline{\alpha} - \overline{\tilde{\alpha}}\right) + \overline{\beta}^u \left(F_t^u - \overbrace{\frac{Cov(F_t^u, F_t^o)}{Var(F_t^o)}}^{\text{OVB}} F_t^o \right)}_{\text{misspecification error}} + \underbrace{n_1^{-1} \sum_{i \in n_1} \varepsilon_{i,t}}_{\text{noise}}$$

The short-and-long consequences of misspecification

$$\overline{AR}_t - \tau_t \approx \underbrace{\left(\overline{\alpha} - \overline{\tilde{\alpha}}\right) + \overline{\beta}^u F_t^u}_{\text{misspecification error}} + \underbrace{n_1^{-1} \sum_{i \in n_1} \varepsilon_{i,t}}_{\text{noise}}$$

- Average noise disappears with many treated firms
- Misspecification error does not
  - Single event: $\overline{AR}_t$ is *stochastic*
- Intercept error: OLS absorbs mean omitted factor premium into $\hat{\alpha}$
  - $\left(\overline{\alpha} - \overline{\tilde{\alpha}}\right) = -\overline{\beta}^u(\mathbb{E}[F_t^u \mid \text{event}])$

$$\overline{AR}_t - \tau_t \approx \overline{\beta}^u(F_t^u - \mathbb{E}[F_t^u \mid \text{pre-period}]) + \underbrace{n_1^{-1} \sum_{i \in n_1} \varepsilon_{i,t}}_{\text{noise}}$$

General framework: setup and notation

- Event timing

$$T_i = \begin{cases} t & \text{if event hits } i \text{ at } t \\ \infty & \text{if never} \end{cases}$$

- Let $C = \{i : T_i = \infty\}$ and S the set of possible event dates.
- Potential returns $R_{i,t}(s)$ if event happens at s, and $R_{i,t}(\infty)$ if never.

$$\tau_i(s,t) = R_{i,t}(s) - R_{i,t}(\infty)$$

$$\tau_{\text{ATT}}(s,t) = \underbrace{\mathbb{E}\left[R_{i,t} \mid T_i = s\right]}_{\text{Observed}} - \underbrace{\mathbb{E}\left[R_{i,t}(\infty) \mid T_i = s\right]}_{\text{Counterfactual Return}}$$

Several estimands of interest

- Can aggregate different ATT into *event-time*

$$\theta_\kappa = \sum_{s \in S} w_s \tau_{\text{ATT}}(s, s+\kappa), \quad w_s = \frac{N_s}{\sum_{s'} N_{s'}} \text{ (under random timing)}$$

- Useful notation for seeing connection to other estimation approaches
- Common special case: cumulative effect $\Theta_H^{\text{CATT}} = \sum_{\kappa=0}^{H} \theta_\kappa$.

Linear factor model assumption for theory results

Assumption: Factor structure

$$\mathbb{E}\left[R_{i,t}(\infty) \mid T_i = s, F_t\right] = \alpha_s + \beta_s F_t,$$

with K common factors F_t and group means (α_s, β_s)

- Explicitly delivers $\mathbb{E}\left[R_{i,t}(0) \mid T_i = s\right]$ used by most event-study models.
- Motivated by finance theory papers but strong e.g. Chamberlain and Rothschild (1983)
- Could generalize factor loadings β_s to account for time-varying characteristics (Daniel, Mota, Rottke, and Santos (2020); Kelly, Pruitt, and Su (2019))

Alternative approaches to counterfactual return

Estimator	Counterfactual	Key assumption
Abnormal returns (CAPM, FF3)	$\hat{\alpha}_i + \hat{\beta}_i F_t^o$ estimated pre-event	Researcher specifies correct factors
Difference-in-means	Average return of control stocks	Treated and control have same avg. loadings
Synthetic control	Weighted portfolio of controls matching pre-event path	Replicating portfolio exists
Gsynth / PCA	PCA-estimated factors + pre-event loadings	Linear factor structure (factors learned from data)

- First two: researcher specifies the model. Last two: data-driven
- Synthetic control and gsynth do not require knowing which factors matter

Synthetic methods use historical paths of realized returns

- Abnormal return estimators specify F_t^o and estimate $\hat{\beta}_i^o$
  - Requires specifying pre-determined factors

Synthetic methods use historical paths of realized returns

- Abnormal return estimators specify F_t^o and estimate $\hat{\beta}_i^o$
  - Requires specifying pre-determined factors

- Synthetic control constructs a *replicating portfolio*
  - Implicitly matches on *all* factor loadings, not just the ones you specify
  - Under a linear factor model, this exactly recovers the true counterfactual

Synthetic methods use historical paths of realized returns

- Abnormal return estimators specify F_t^o and estimate $\hat{\beta}_i^o$
  - Requires specifying pre-determined factors

- Synthetic control constructs a *replicating portfolio*
  - Implicitly matches on *all* factor loadings, not just the ones you specify
  - Under a linear factor model, this exactly recovers the true counterfactual

- Gsynth uses PCA to *learn* the factor structure from control stocks
  - Cross-validates the number of factors
  - Then estimates treated loadings from pre-event data

- GSynth and synth are consistent as $n_0, T \to \infty$ [formal results]

When does bias matter? Three cases

- **Case 1: Many events, random timing, short horizon** ✓ Standard methods work

$$\overline{AR}_t - \tau_t \approx \overline{\beta}^u (E[F_t^u] - \mathbb{E}[F_t^u \mid \text{pre-period}])$$

  - Priced premium absorbed by the intercept (α_i), unpriced premia averaged out
  - This is why "the model doesn't matter" (Brown & Warner (1985), Shleifer (1986))

When does bias matter? Three cases

- **Case 1: Many events, random timing, short horizon** ✓ Standard methods work

$$\overline{AR}_t - \tau_t \approx \overline{\beta}^u (E[F_t^u] - \mathbb{E}[F_t^u \mid \text{pre-period}])$$

  - Priced premium absorbed by the intercept (α_i), unpriced premia averaged out
  - This is why "the model doesn't matter" (Brown & Warner (1985), Shleifer (1986))
- **Case 2: Single event or extreme volatility** × Standard methods fail

$$\overline{AR}_t - \tau_t \approx \overline{\beta}^u (F_t^u - \mathbb{E}[F_t^u \mid \text{pre-period}])$$

  - No averaging — bias $\approx \overline{\beta}^u F_t^u$ (plus a loading-gap term)
  - Crisis day: small loading gap $\times$ huge factor realization $=$ large bias

When does bias matter? Three cases

- **Case 1: Many events, random timing, short horizon** ✓ Standard methods work

$$\overline{AR}_t - \tau_t \approx \overline{\beta}^u (E[F_t^u] - \mathbb{E}[F_t^u \mid \text{pre-period}])$$

  - Priced premium absorbed by the intercept (α_i), unpriced premia averaged out
  - This is why "the model doesn't matter" (Brown & Warner (1985), Shleifer (1986))
- **Case 2: Single event or extreme volatility** × Standard methods fail

$$\overline{AR}_t - \tau_t \approx \overline{\beta}^u (F_t^u - \mathbb{E}[F_t^u \mid \text{pre-period}])$$

  - No averaging — bias $\approx \overline{\beta}^u F_t^u$ (plus a loading-gap term)
  - Crisis day: small loading gap $\times$ huge factor realization $=$ large bias
- **Case 3: Long horizon** × Standard methods fail

$$\hat{\Theta}_H - \Theta_H \approx \overline{\beta}^u \sum_{\kappa=0}^{H} (F_{s+\kappa}^u - \mathbb{E}[F^u \mid \text{pre-period}])$$

  - Event window not representative: $\mathbb{E}(F^u \mid \text{post}) \neq \mathbb{E}(F^u \mid \text{pre})$
  - Even "small" daily bias *compounds*: 0.02%/day → 5% over one year

Synthetic control and gsynth: consistent in all three cases [formal results]

What about calendar-time portfolios?

- Calendar-time portfolio (Fama (1998)): form portfolio of firms in event window, regress on factors, estimate α
- We show calendar-time α is a *reweighted* version of event-time ATT
  - Downweights event-clustered periods; coincides with event-time only if event flow is uniform
  - Same bias structure as event-time — same three cases apply

What about calendar-time portfolios?

- Calendar-time portfolio (Fama (1998)): form portfolio of firms in event window, regress on factors, estimate α
- We show calendar-time α is a *reweighted* version of event-time ATT
  - Downweights event-clustered periods; coincides with event-time only if event flow is uniform
  - Same bias structure as event-time — same three cases apply
- One key distinction: when treatment *changes* factor loadings (e.g., M&A raises beta)
  - Event-time ATT $=$ total return impact
  - Calendar-time $\alpha =$ risk-adjusted anomaly (relative to new loadings)
  - Neither is inherently correct — depends on research question [details]

Buy-and-hold abnormal returns are hard

- Geometric (BHAR) ATT differs from arithmetic (CAR) ATT:

$$\Theta_H^{\text{geo,ATT}} \approx \Theta_H^{\text{ATT}} - \sum_{\kappa=0}^{H} \mathbb{E}\left[R_{i,s+\kappa}(\infty)\tau_i(s, s+\kappa) + \frac{1}{2}\tau_i(s, s+\kappa)^2 \mid T_i = s \right]$$

- Even a portfolio with the *correct* expected return is a bad counterfactual for BHAR if it has different variance (volatility drag)
- Recommendation: focus on arithmetic ATT for tractability and unbiasedness

Key takeaway from the theory

1. Short-run event studies with many randomly-timed events are robust
2. Single-event studies during volatile periods are fragile
3. Long-run event studies are almost always biased
4. When in doubt, report synth or gsynth alongside standard methods
5. Do cumulative arithmetic returns over buy-and-hold returns

Application 1: Geithner

Empirical example 1: Acemoglu, Johnson, Kermani, Kwak, Mitton

- Acemoglu et al. (2016) study how the leak of Timothy Geithner's nomination as U.S. Treasury Secretary on Nov 21, 2008 affected firms connected to him
  - Focus on pooled average treatment effect (ATT) for five methods: abnormal returns, synthetic control, gsynth and synthdid
- Paper compares *within* banks connected vs. not, we expand control group
- Key features:
  - Single event
  - Unusual timing (financial crisis)
  - Short horizon

Contents lists available at ScienceDirect

Journal of Financial Economics

journal homepage: www.elsevier.com/locate/jfec

The value of connections in turbulent times: Evidence from the United States☆

Daron Acemoglu[a], Simon Johnson[b], Amir Kermani[c], James Kwak[d], Todd Mitton[e,*]

[a] MIT Department of Economics, 77 Massachusetts Avenue Building E52, Room 446, Cambridge, MA 02142-1347, United States
[b] MIT Sloan School of Management, 50 Memorial Drive E62-420, Cambridge, MA 02142, United States
[c] Haas School of Business, University of California, Berkeley 545 Student Services Building #1900, Berkeley, CA 94720-1900, United States
[d] University of Connecticut School of Law, 55 Elizabeth Street Hosmer 118, Hartford, CT 06105-2290, United States
[e] Brigham Young University, Marriott School of Management, 640 TNRB, Provo, UT 84602, United States

ARTICLE INFO

Article history:
Received 28 May 2015
Revised 14 September 2015
Accepted 12 October 2015
Available online 27 April 2016

JEL Classification:
G01
G14
G21
G28

Keywords:
Political connections
Economic crises
Institutions

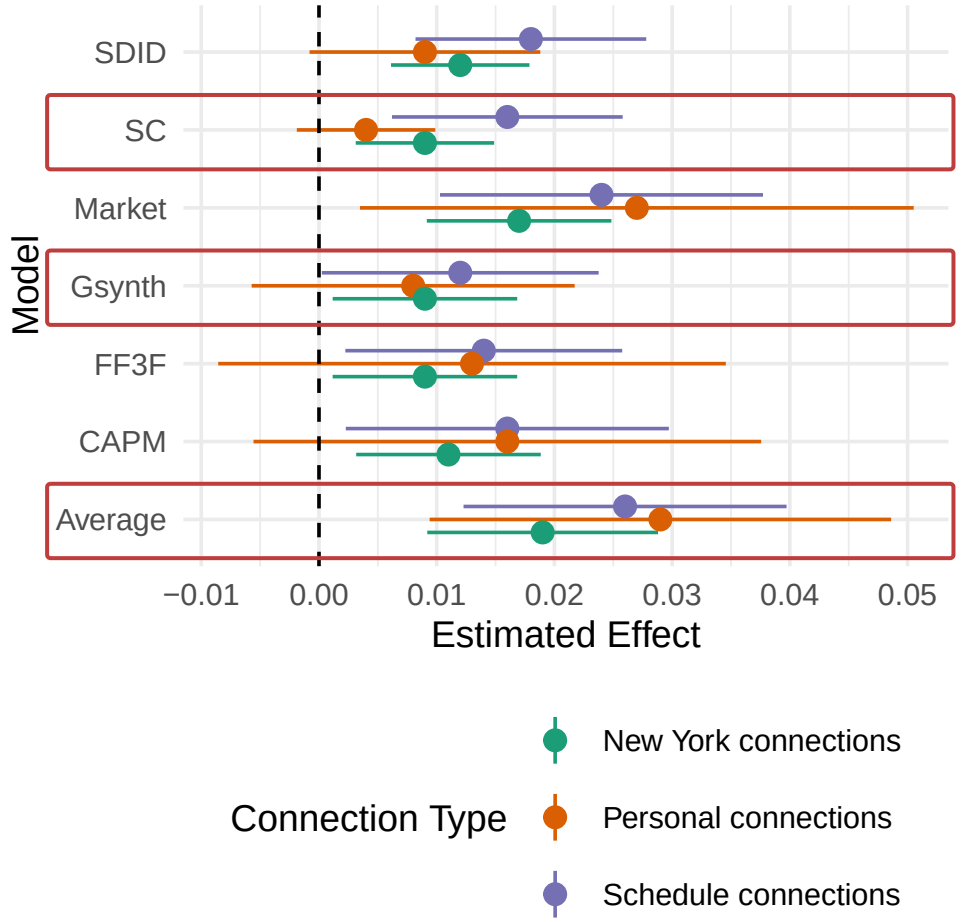
ABSTRACT

The announcement of Timothy Geithner as nominee for Treasury Secretary in November 2008 produced a cumulative abnormal return for financial firms with which he had a prior connection. This return was about 6% after the first full day of trading and about 12% after ten trading days. There were subsequently abnormal negative returns for connected firms when news broke that Geithner's confirmation might be derailed by tax issues. Personal connections to top executive branch officials can matter greatly even in a country with strong overall institutions, at least during a time of acute financial crisis and heightened policy discretion.

© 2016 Elsevier B.V. All rights reserved.

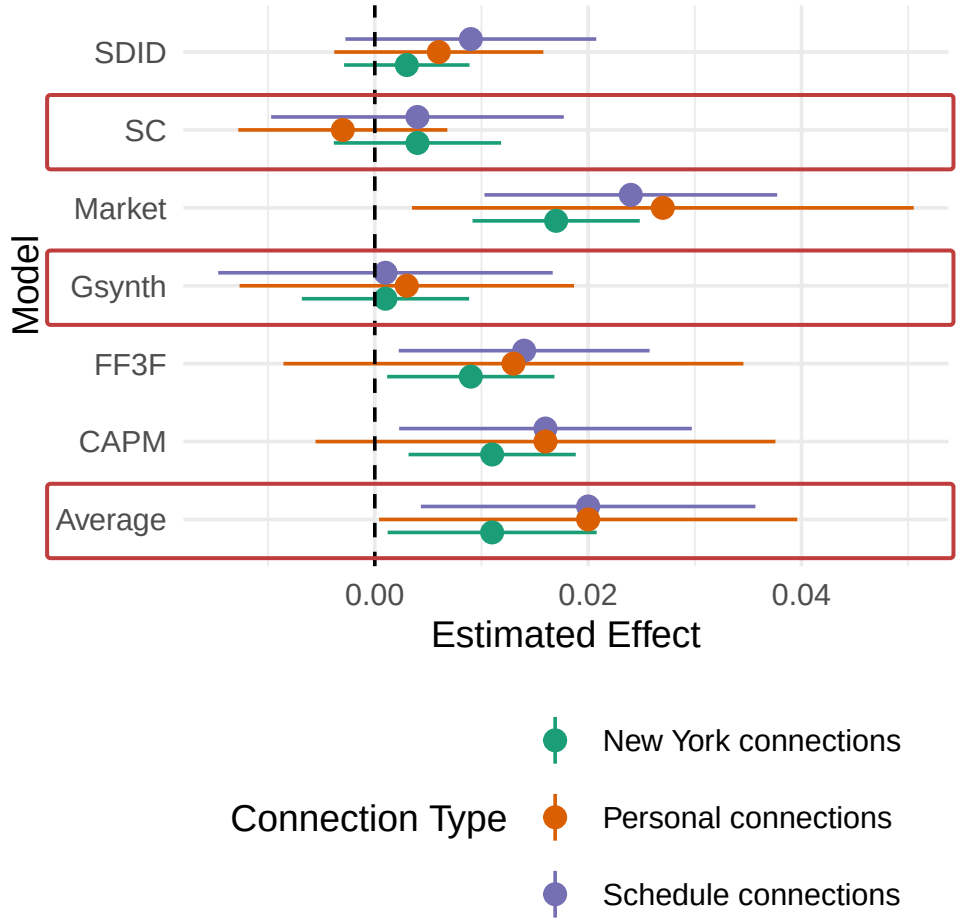
Results are much closer to zero using synthetic methods

- Effect sizes are large in the paper, shrink significantly with synthetic approach, even when focused on just banks as controls

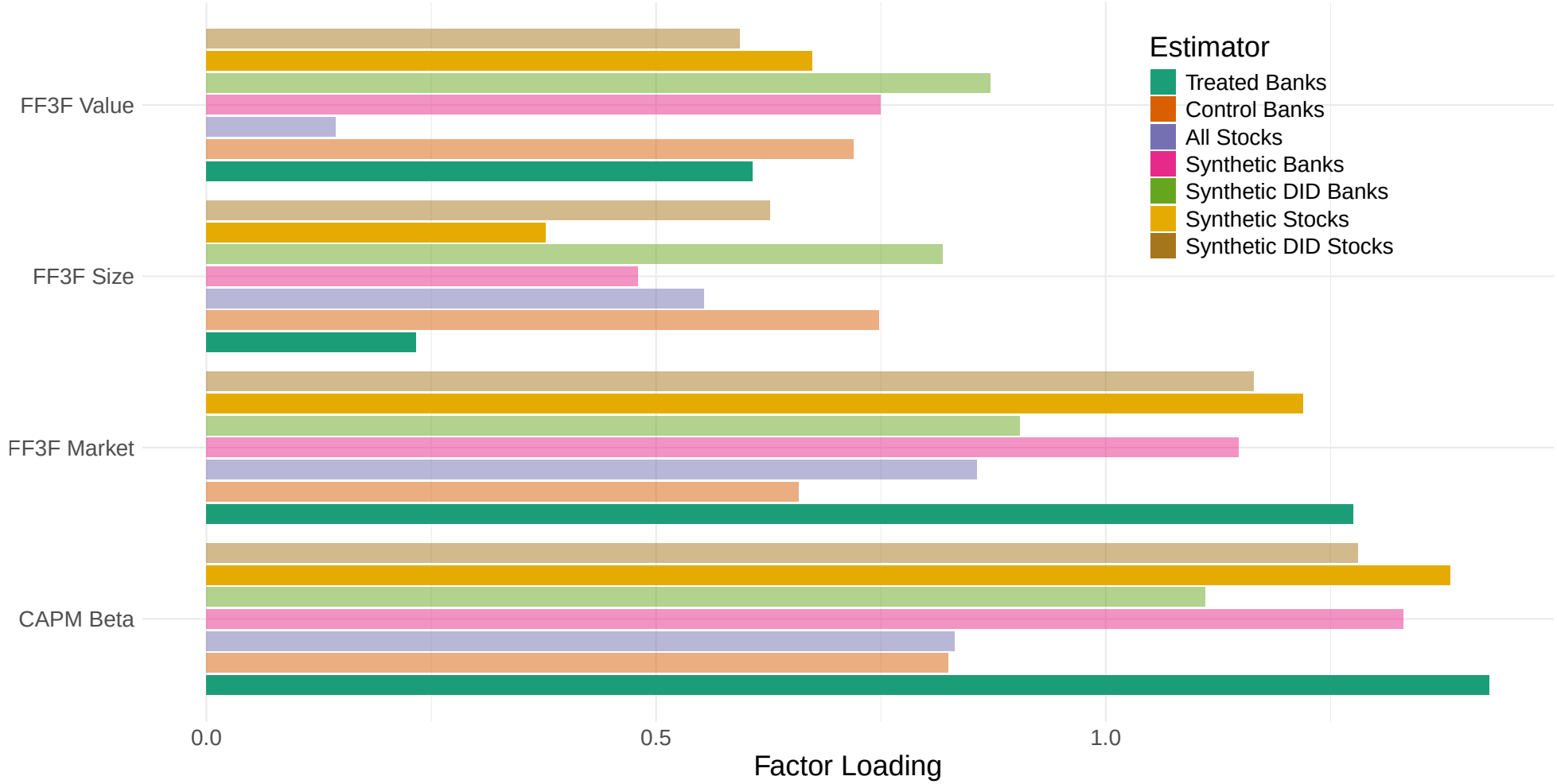


Results are much closer to zero using synthetic methods

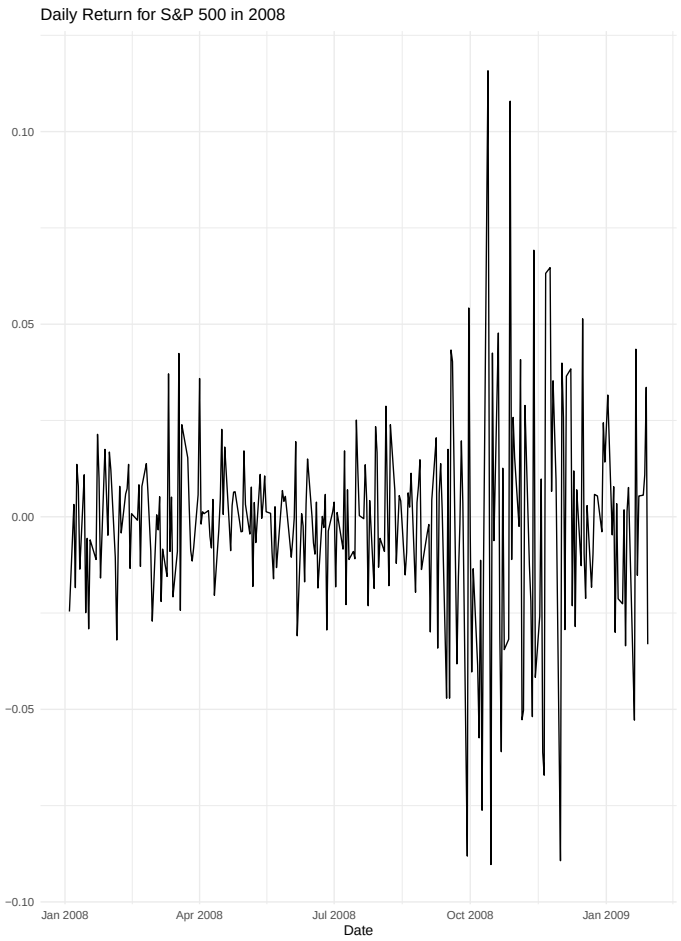
- Effect sizes are large in the paper, shrink significantly with synthetic approach, even when focused on just banks as controls
- When expanded to the full universe of control firms, all estimated effects are effectively zero
  - Why?



Reason 1: differences in factor loadings



Reason 2: non-random timing



- Timing of event is correlated with significant risk factors
- S&P surged on Nov 21 — a market move in the extreme tail of its distribution
- Beta gap of $0.6 \times 6.9\%$ market move $\approx 4.1\%$ mechanical bias (one day)

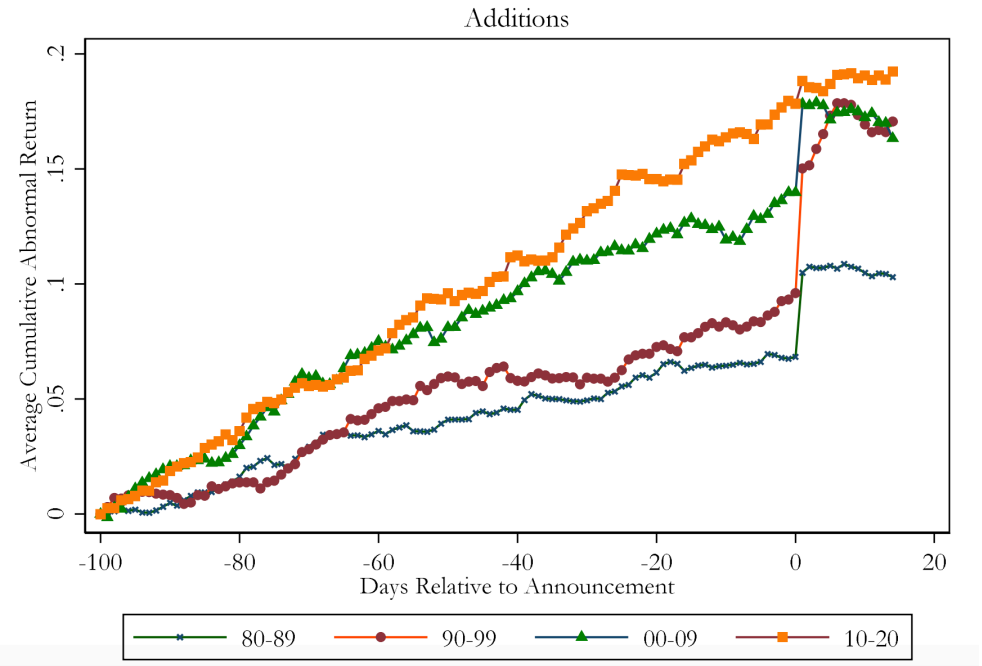
Geithner takeaway

- With a **single event** during **extreme volatility**, the standard approach can generate large spurious effects
- A beta gap of 0.6 $\times$ a 6.9% market day $=$ 4.1% mechanical bias *on one day*
- Synthetic methods eliminate the apparent effect entirely
- For single-event studies, check for balance using factor loading comparisons and synthetic control estimates

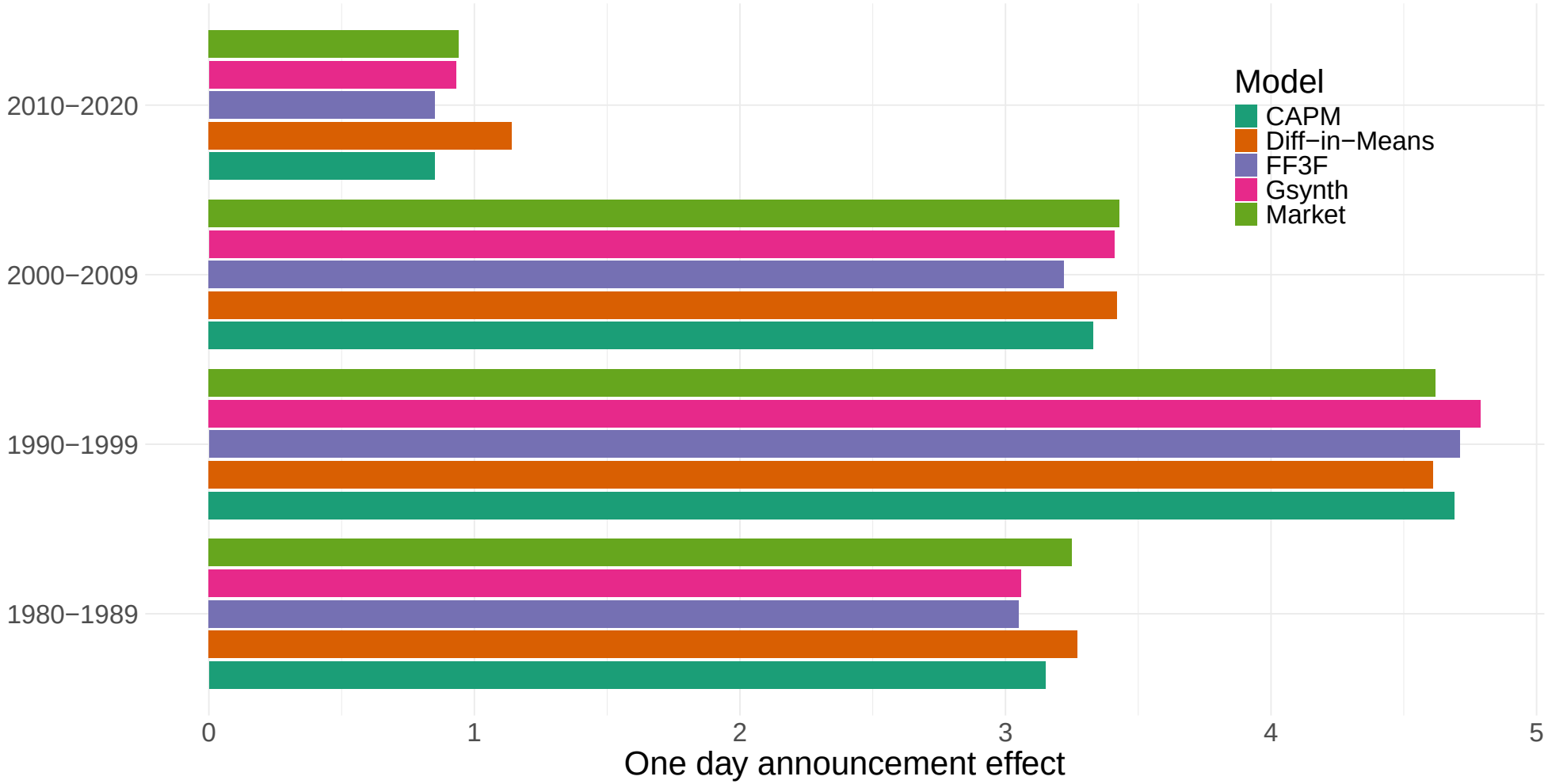
Application 2: S&P 500 Index Inclusion

Empirical example 2: S&P Index Inclusion Effect

- S&P 500 index inclusion effect: firms added to the index experience a large positive return on the day of inclusion
- Replicate analysis from Greenwood and Sammon (2025)
  - S&P inclusions from 1976–2020
  - Use announcement dates from Siblis Research, if missing, use day prior to effective day
- Key features:
  - Many events
  - As-if random timing [evidence]
  - Short- and long- horizon

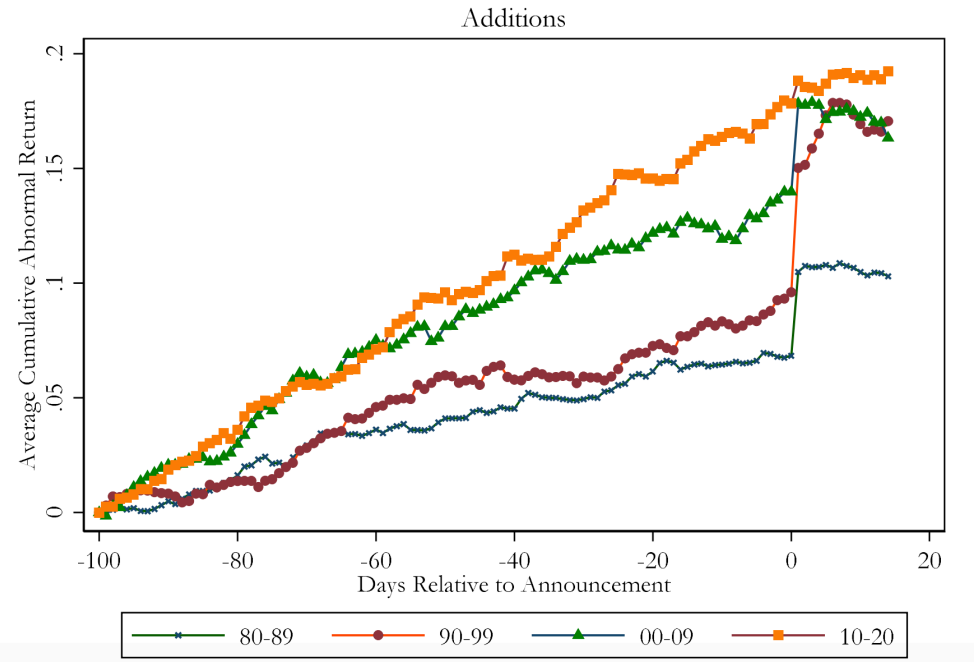


Method for short-run estimation does not matter

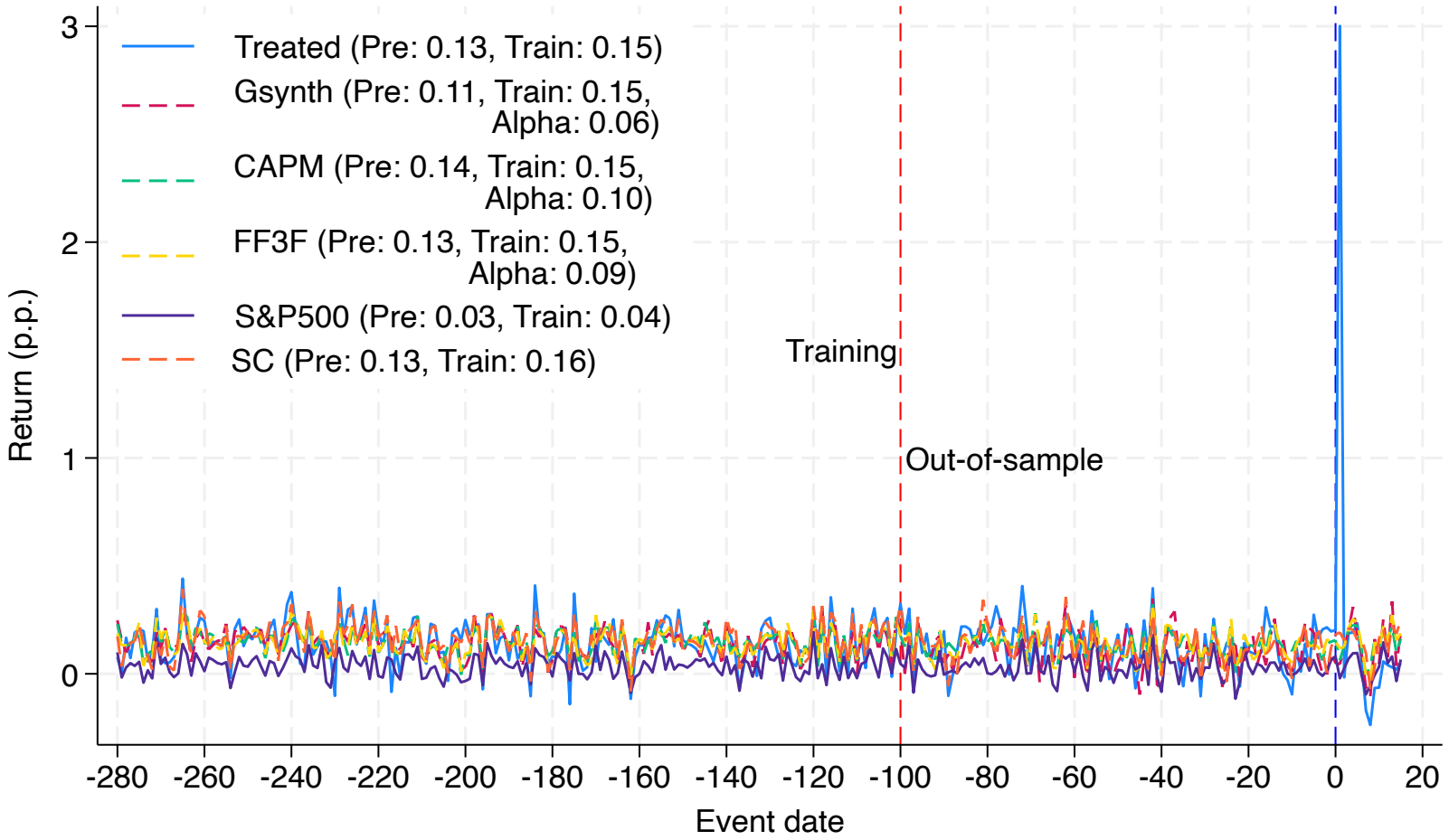


Pre-inclusion drift as a long-run effect

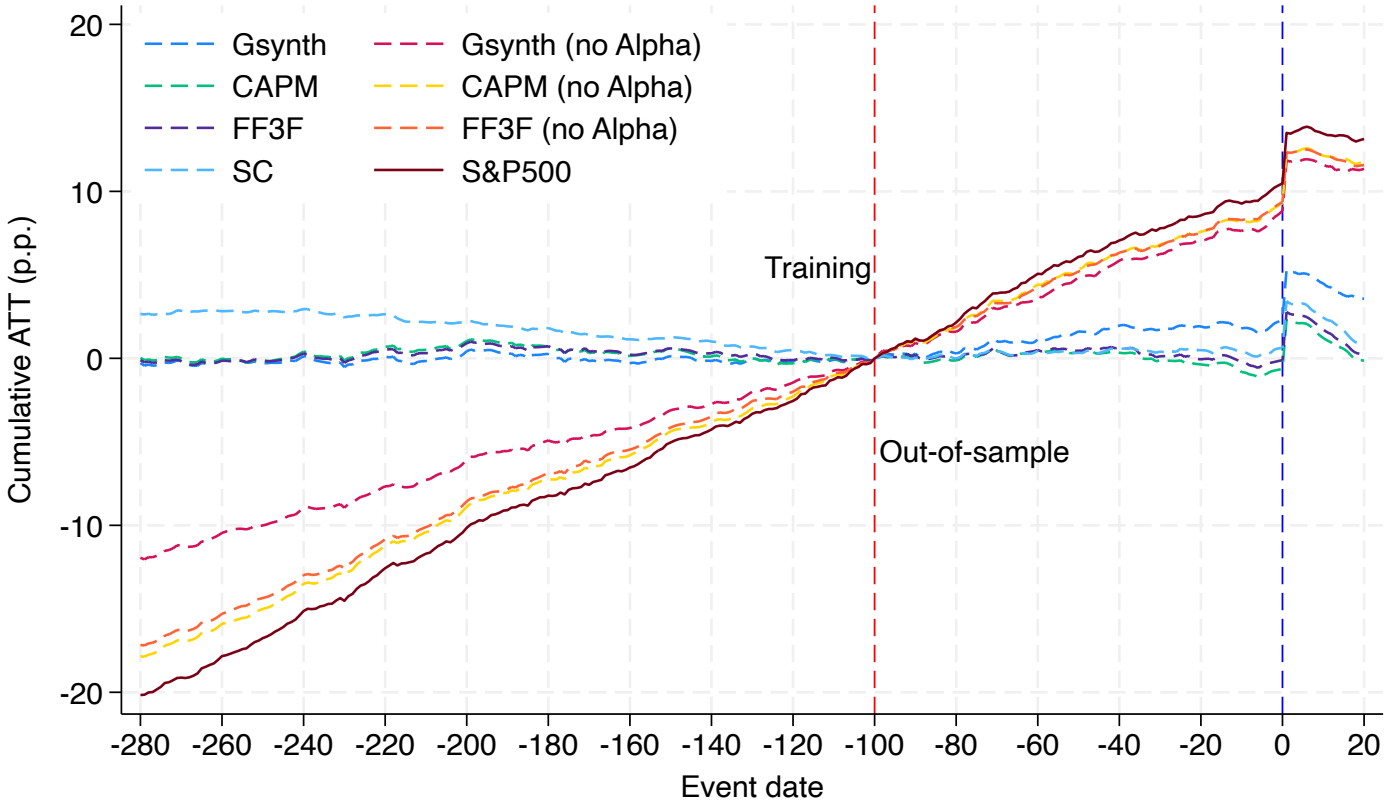
- We study the pre-inclusion "drift" as a form of long-run bias
- Often, drift is pointed to as a puzzle, evidence of potential front-running, or other market activity
- But: firms included in S&P 500 are *different* from other firms
  - Alpha in factor models is large and positive even 280 days pre-event
  - Reflects model misspecification: $\hat{\alpha} = \alpha_{\text{true}} + \beta^{\text{unobs}} \mathbb{E}\left(F^{\text{unobs}}\right)$



Per-Period ATTs for Index Inclusion



Factor methods remove the drift — only with estimated α

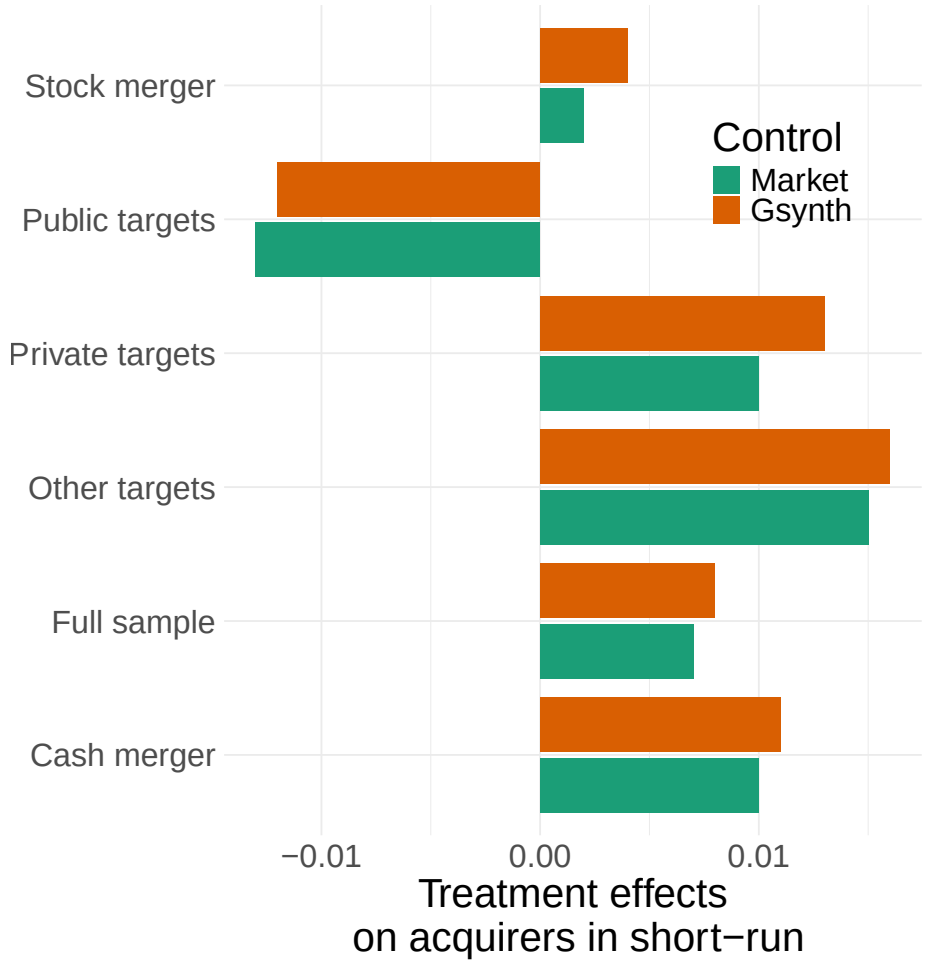


Without $\hat{\alpha}$, CAPM / FF3F / gsynth leave the drift; synthetic control (no intercept) matches the pre-period on its own.

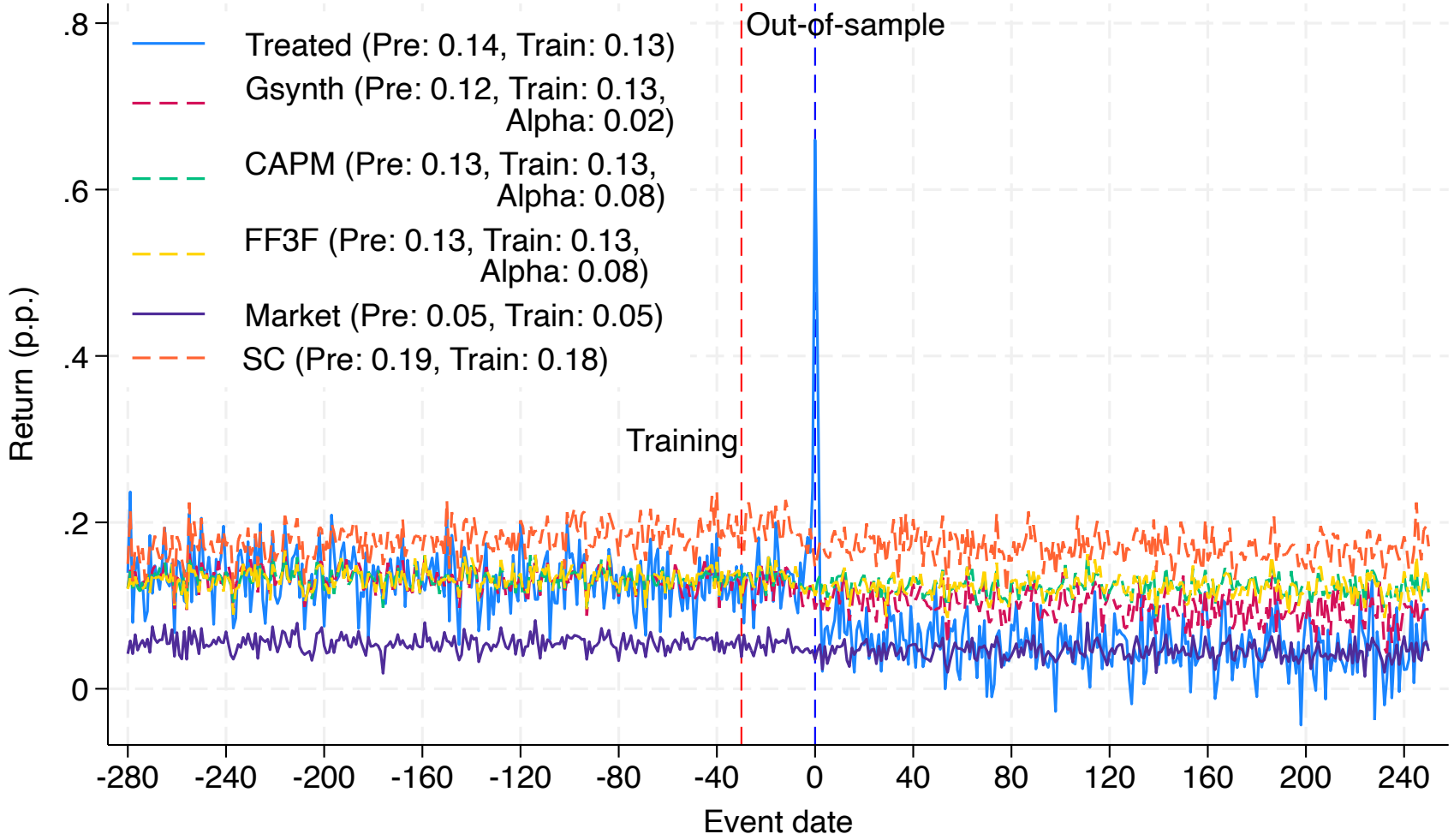
Application 3: Mergers & Acquisitions

Empirical example 3: M&A

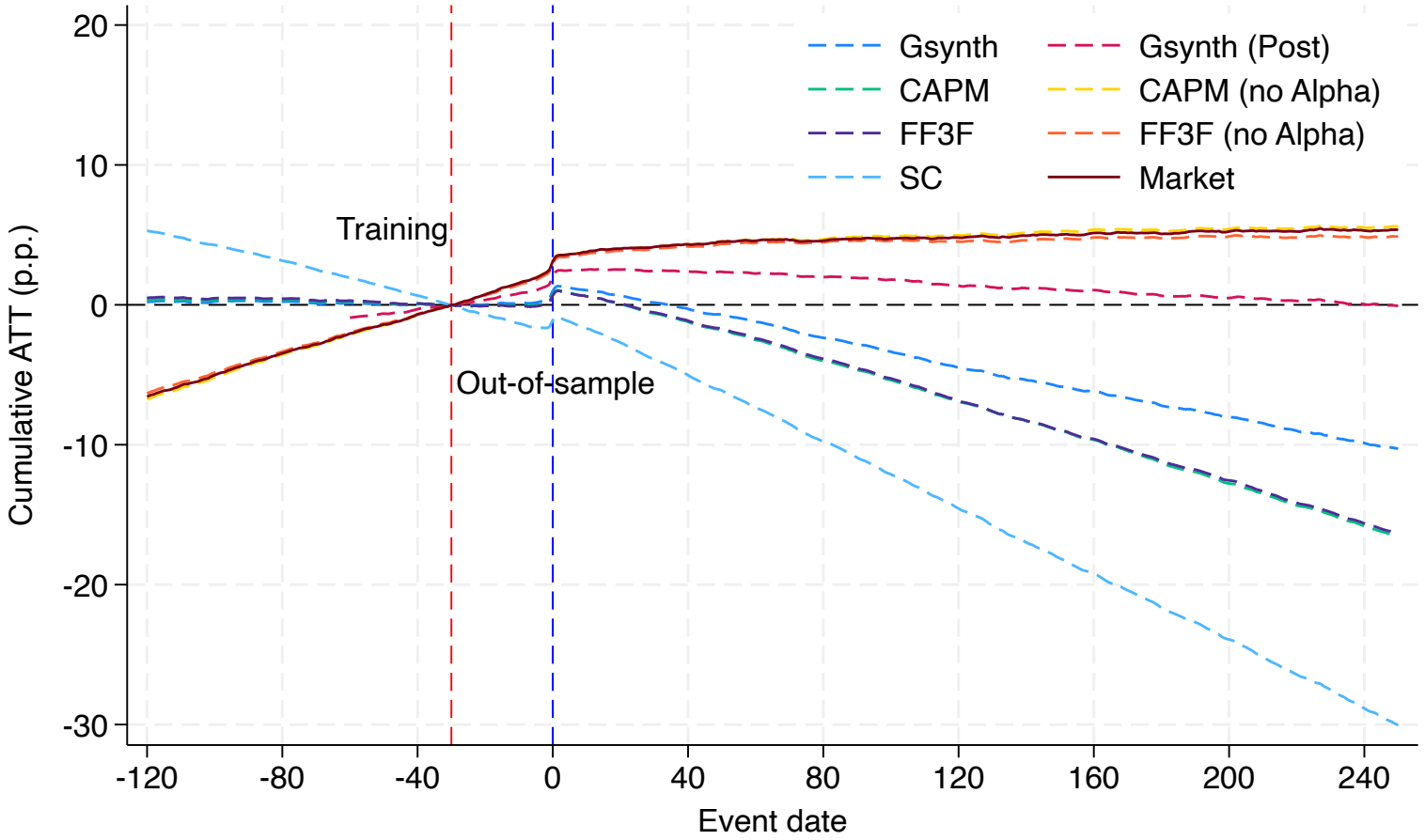
- What is impact of acquisitions on *acquirer* returns?
- Replicate analysis from Malmendier (2017)
  - All acquisitions in SDC from 1980–2024
- Key features:
  - Many events
  - Quasi-random timing [evidence]
  - Short- and long- horizon
- Short-run effects quite similar, but...



Per-Period ATTs for M&A



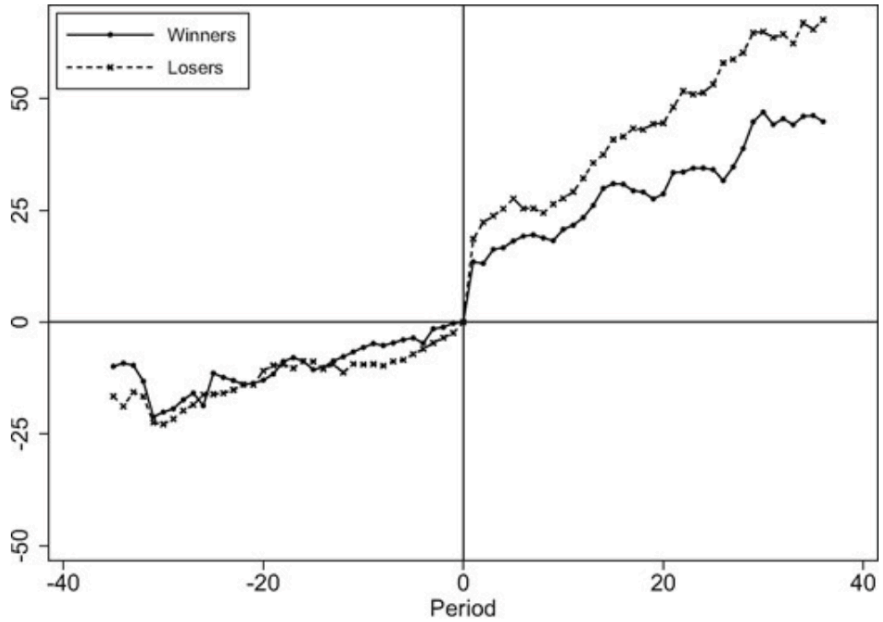
Synthetic methods match on overpricing in pre-period



Application 4: Close Merger Contests

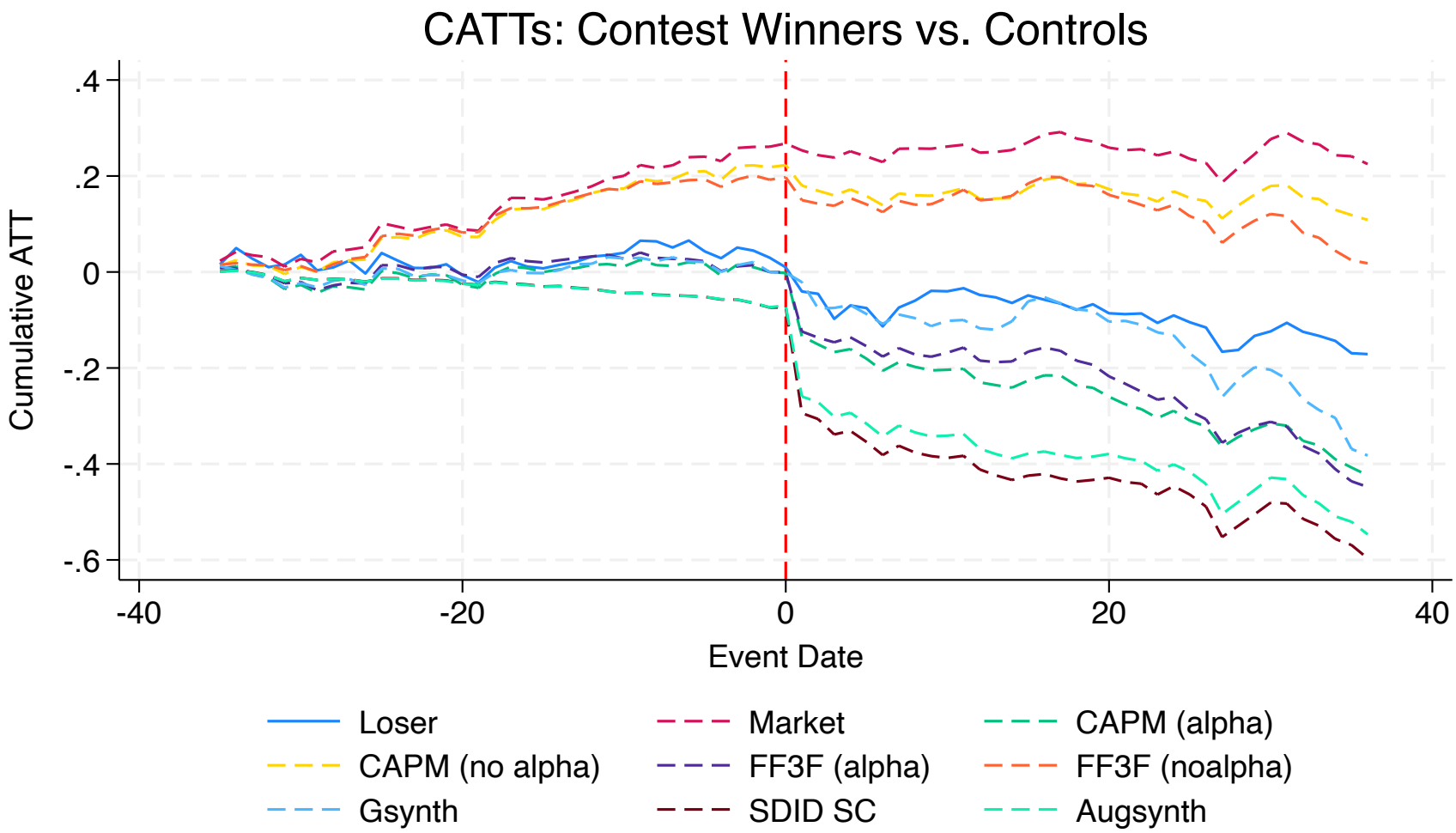
Close merger contests as design-based comparison

- Malmendier, Moretti, and Peters (2018) use "close" mergers — bidding contests
  - Goal is to find as-if coinflip mergers
  - Many events (100 different mergers)
  - Short- and long- horizon
- Striking takeaway in paper is losers doing better
- Compare model based approach to design-based approach
  - LaLonde-style exercise: contest losers serve as quasi-experimental counterfactual



(d) Unadjusted buy-and-hold returns

Design-based approach gives very different answers



Key lessons from the close contest comparison

- The design-based loser counterfactual (benchmark truth) shows:
  - Smallest, least-trending pre-period
  - Small negative post-contest ATT — consistent with mild harm to acquirers
- Model-based approaches:
  - **Gsynth**: closest to loser counterfactual
  - **Synthetic control**: larger declines during and after contest
  - **CAPM, FF3F**: fit well pre-period but predict severe long-run declines not supported by loser comparison
- For long-run analysis: quasi-experimental variation is likely superior to model-based approaches

Conclusion

What should you take away from this paper?

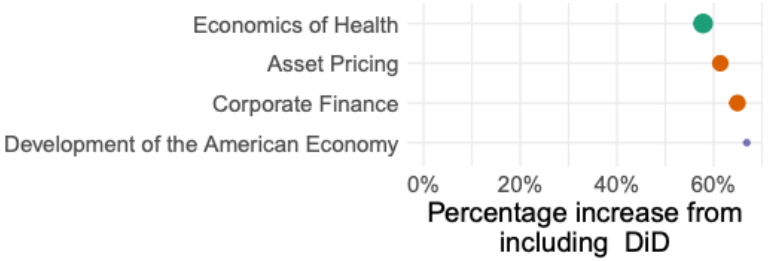
- The “model doesn’t matter” folk wisdom is right — but only for short-run, many-event, random-timing studies

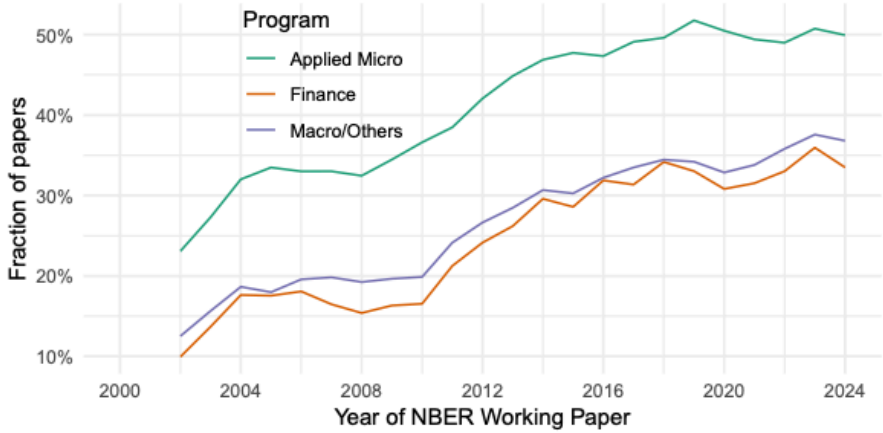
> *The results were not materially different when returns were not corrected for market movements.* — Shleifer (1986)

- It is wrong for:
  - Single events during volatile periods (Geithner)
  - Long-horizon analyses (M&A, pre-inclusion drift)
  - Settings where treated firms have distinctive factor loadings
- The core issue: asset pricing models predict *expected* returns well, but not *contemporaneous* returns — and counterfactual prediction needs the latter

Financial event studies are a causal-inference tool

- Same biases affect panel diff-in-diff studies
  - But asset prices incorporate information far faster than other outcomes
  - Better motivation for factor models in outcomes
- Finance has lagged on causal-inference tools
  - Yet owns experiments and outcomes other fields lack





(a) Identification

Goldsmith-Pinkham (2024)

Appendix

Formal bias results: single event

As $n_s, n_c, T \to \infty$:

$$\tau^{AR}(s,t) - \tau_{\text{ATT}}(s,t) \xrightarrow{p} (\alpha_s - \tilde{\alpha}_s) + \left(\beta_s F_t - \tilde{\beta}_s F_t^o\right)$$

$$\hat{\tau}^{\text{cont}}(s,t) - \tau_{\text{ATT}}(s,t) \xrightarrow{p} (\alpha_s - \alpha_\infty) + (\beta_s - \beta_\infty) F_t$$

$$\hat{\tau}^{\text{gsynth}}(s,t) - \tau_{\text{ATT}}(s,t) \xrightarrow{p} 0$$

$$\hat{\tau}^{\text{SC}}(s,t) - \tau_{\text{ATT}}(s,t) \xrightarrow{p} 0$$

- Even with $n_s, n_c, T \to \infty$, AR and DiD are biased if the factor model is wrong
- Synthetic control and gsynth are consistent

Formal bias results: multiple staggered events

Assume $n_s, n_c, T \to \infty$ and each date in $\mathcal{S}$ has non-trivial treatment probability. Then

$$\hat{\theta}^{AR}_{\kappa} - \theta^{\text{ATT}}_{\kappa} \xrightarrow{p} \mathbb{E}\Big[(\alpha_s - \tilde{\alpha}_s) + \Big(\beta_s F_{s+\kappa} - \tilde{\beta}_s F^o_{s+\kappa}\Big) \mid T_i \in S\Big]$$

$$\hat{\theta}^{\text{gsynth}}_{\kappa} - \theta^{\text{ATT}}_{\kappa} \xrightarrow{p} 0 \qquad \hat{\theta}^{\text{SC}}_{\kappa} - \theta^{\text{ATT}}_{\kappa} \xrightarrow{p} 0$$

Under random timing with stationary factors:

- The OLS intercept absorbs mean factor premium: $\alpha_i - \tilde{\alpha}_i = -\beta_i \mathbb{E}[F_t] + \tilde{\beta}_i \mathbb{E}[F^o_t]$
- This cancels bias exactly → theoretical foundation for “the model doesn’t matter”

Under random timing with **non-stationary** factors: bias remains $\propto \mathbb{E}(\beta_i)(\mathbb{E}[F_t \mid \text{post}] - \mathbb{E}[F_t \mid \text{pre}])$

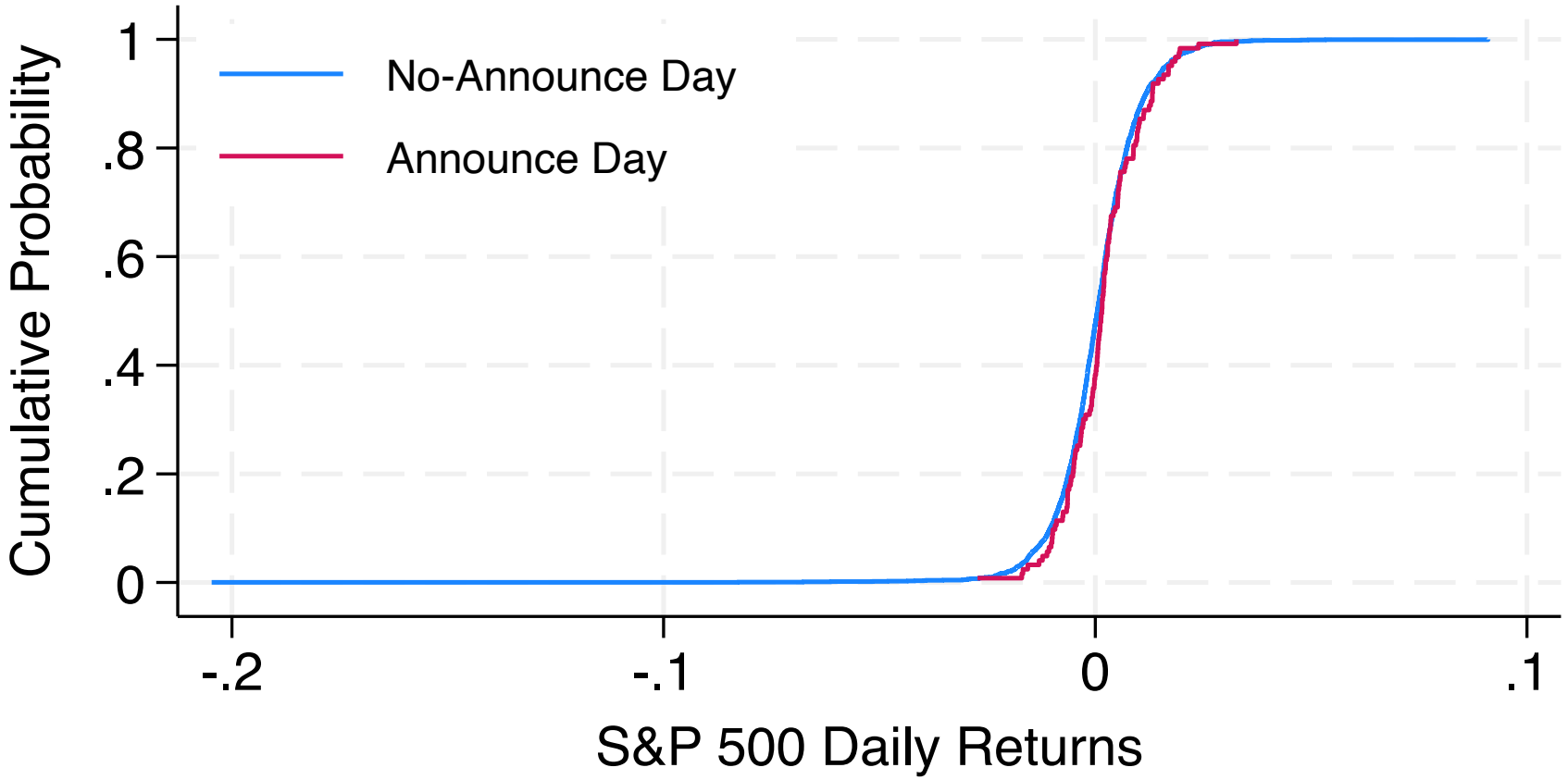
Timing propensity score

$$p_t(X_i, F) = \Pr(T_i = t \mid X_i, F), \quad X_i = (\alpha_i, \beta_i)$$

- **Random assignment:** $p_t(X_i, F) = p_t(F)$
- **Random timing:** $p_t(X_i, F) = p_t(X_i)$

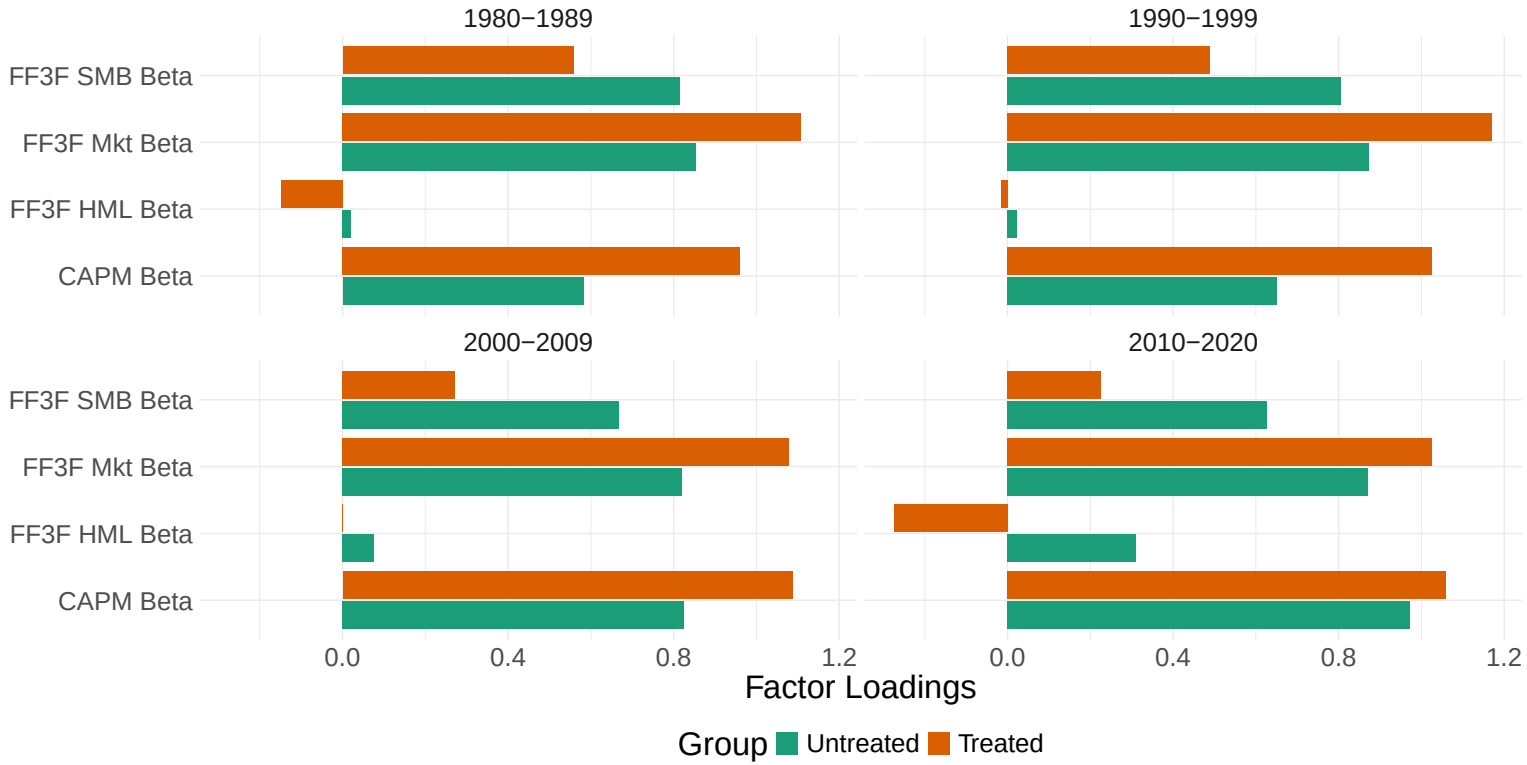
Random assignment controls who is treated; random timing controls *when*.

One-day event effect is roughly consistent because of random timing

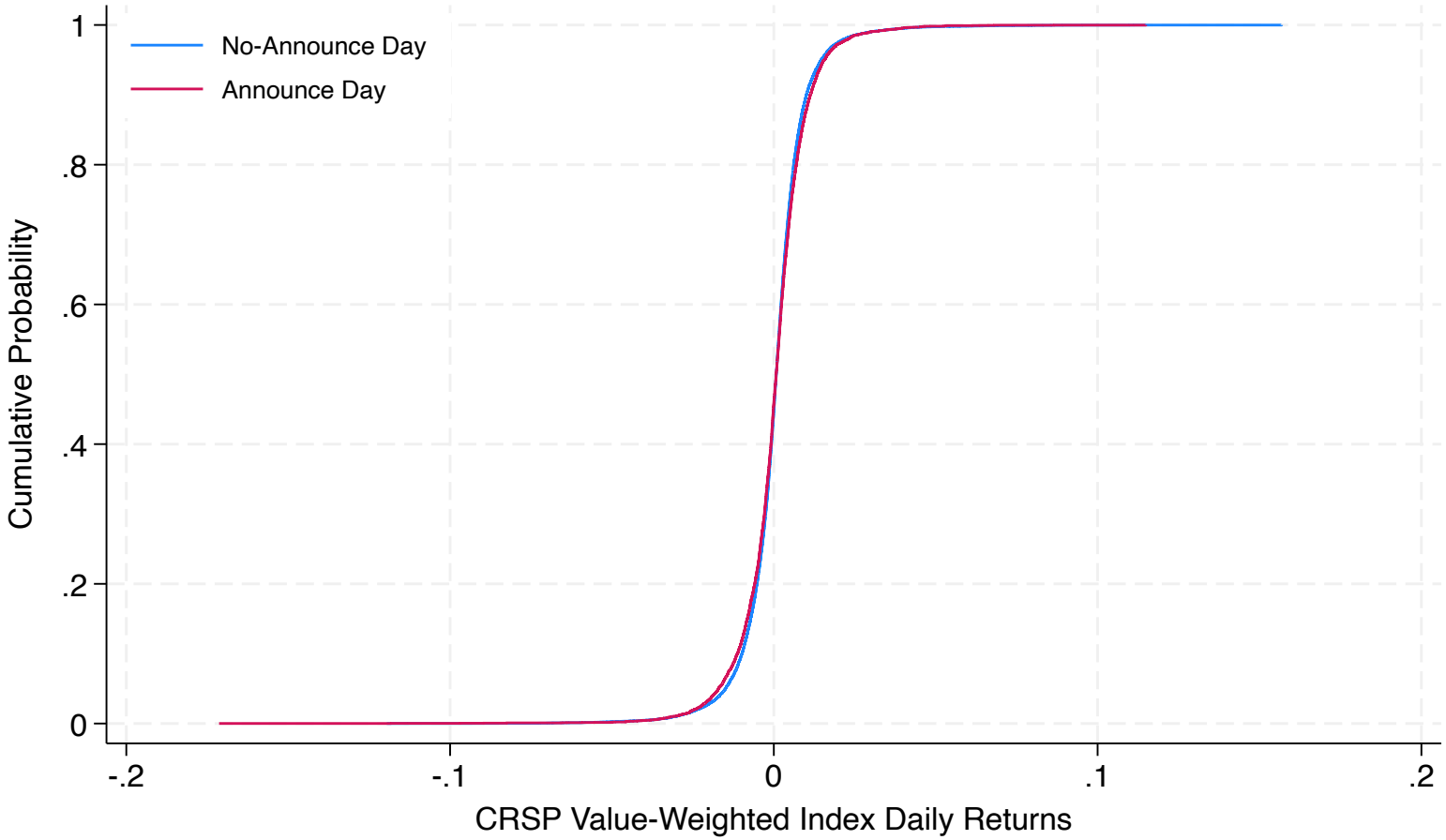


However, not randomly assigned to firms

- Treated firms are significantly different than untreated firms



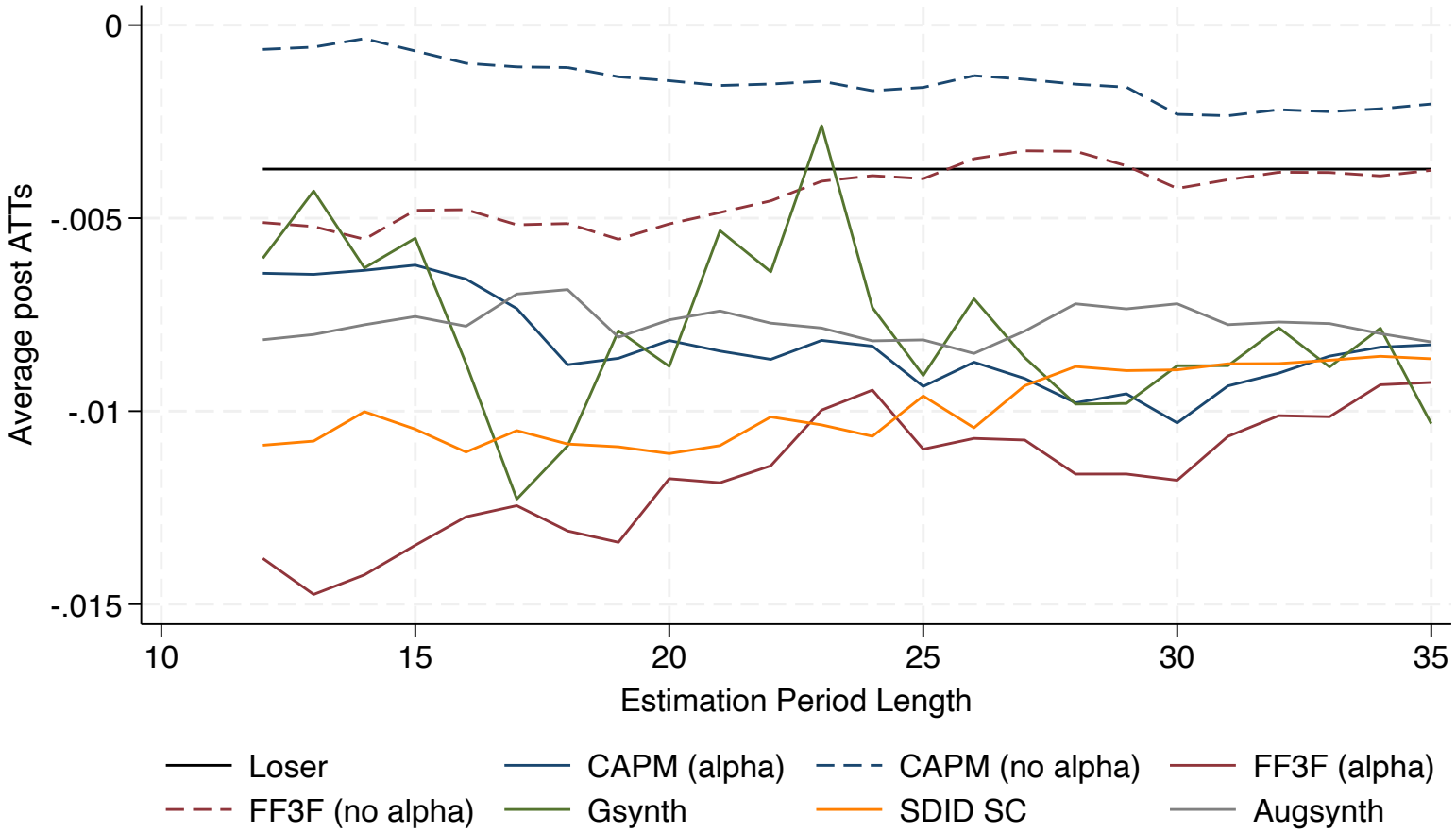
Short-term effects are similar because of random timing



Individual firm estimates: noisy but not necessarily biased

- For a single treated firm, three sources of randomness:
  1. Estimated parameters ($\hat{\alpha}_i, \hat{\beta}_i$)
  2. Factor realizations (F_t)
  3. Idiosyncratic error (ε_{it})
- With many firms, ε_{it} averages away; with one firm it does not
  - Relevant for litigation/damages context (Baker (2020))
- With many individual treatment effects (e.g. patent issuance, Kogan et al. (2017)):
  - Averages follow the same bias patterns as cohort estimates
  - Shrinkage estimators or portfolio groupings can help

Sensitivity to estimation window length (close contests)



Extension: Testing for over- and underreaction

- Common test: regress long-run returns on short-run returns

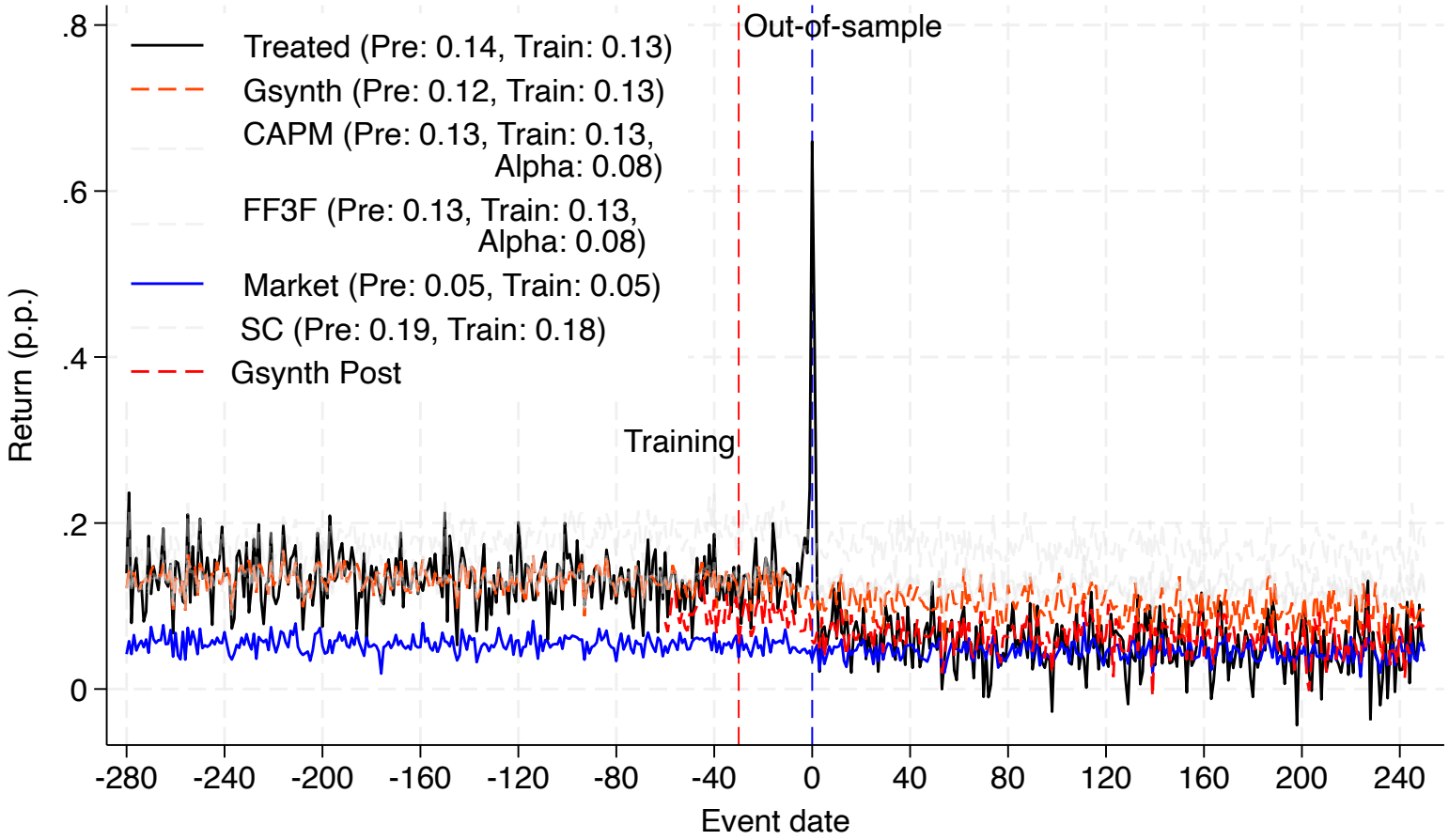
$$\tau_i^{\text{long}} = \alpha + \gamma \cdot \tau_i^{\text{short}} + \varepsilon_i$$

- $\gamma = 0$: efficient; $\gamma > 0$: underreaction; $\gamma < 0$: overreaction
- Two problems under misspecification:
  1. Measurement error in $\hat{\tau}^{\text{short}}$ attenuates $\hat{\gamma}$ toward zero
  2. Spurious positive correlation from common factor loading errors:

$$Cov\left(\widehat{CAR}_i^{\text{short}}, \widehat{CAR}_i^{\text{long}}\right) = Cov\left(\tau_i^{\text{short}}, \tau_i^{\text{long}}\right) + \left(\beta_i - \tilde{\beta}_i\right)^2 \cdot \Sigma_F$$

- Buy-and-hold can generate spurious overreaction: higher-volatility firms → larger volatility drag

Per-Period ATTs for M&A (Gsynth Pre vs Post Period)



What goes wrong in finite samples

- SC minimizes *realized* pre-period fit (Ferman & Pinto (2021))
  - With many donors and a short window, a good fit can *conceal* imbalance
- Let $s_t = \sum_j w_j R_{jt}$ be the synthetic path and $e_t = R_{1t} - s_t$ the pre-period residual
- Moving weight α toward donor j changes the fitting loss (mean squared e_t) by

$$\Delta L = \underbrace{-2\alpha \cdot \overline{(R_{1t} - s_t)(R_{jt} - s_t)}}_{\text{reward: donor tracks unexplained residual}} + \underbrace{\alpha^2 \cdot \overline{(R_{jt} - s_t)^2}}_{\text{cost: donor deviates from current synthetic}}$$

- R_{1t} and R_{jt} contain both the factor and idiosyncratic noise
  - Overfit spread $\propto \sigma_j \sigma_1 / \sqrt{T_{\text{pre}}}$: noisy donors supply the most extreme matches
- $\mathbb{E}\left[e_t (R_{jt} - s_t)\right] = 0$ under the null, but not in finite T!

A finance-specific wrinkle: noisy prices inflate measured returns

- Stationary, mean-zero transitory error η_{it} with autocovariance $\gamma_i(k) = \mathrm{Cov}\big(\eta_{it}, \eta_{i,t-k}\big)$:

$$P^*_{it} = P_{it}(1+\eta_{it}) \quad \Rightarrow \quad \mathbb{E}[R^*_{it}] = \mathbb{E}[R_{it}] + m_i, \quad m_i = \gamma_i(0) - \gamma_i(1) \geq 0$$

  - Expand $R^*_{it} \approx R_{it} + \eta_{it} - \eta_{i,t-1} - \eta_{i,t-1}\eta_{it} + \eta^2_{i,t-1}$ and take expectations: linear terms vanish, $\mathbb{E}\big[\eta^2_{i,t-1}\big] = \gamma_i(0)$, $\mathbb{E}\big[\eta_{i,t-1}\eta_{it}\big] = \gamma_i(1)$

  (Blume & Stambaugh (1983); Roll (1984); Asparouhova, Bessembinder & Kalcheva (2013))

- m_i is a *persistent* level difference in expected measured returns
- Under the null with perfect factor matching, the per-period bias of SC is

$$\mathbb{E}[\hat{\tau}_t] = m_1 - \sum_j w_j m_j = \underbrace{(m_1 - \overline{m})}_{\text{assignment}} - \underbrace{\left(\sum_j w_j m_j - \overline{m}\right)}_{\text{estimator selection}}$$

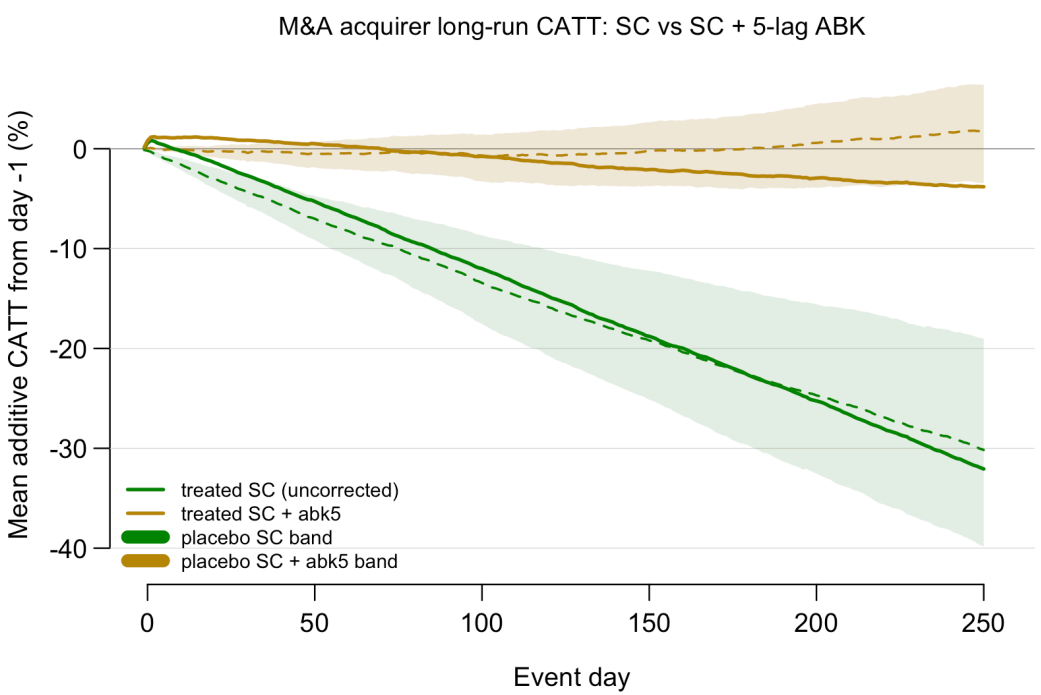
- The estimator manufactures the bias even in an RCT
  - Targeting realized path vs. factor loadings creates issue

Possible Solutions

- k-lag ABK weighting
  - Daily rebalancing resets each donor to its target weight every day, so a price with bid-ask bounce is bought low and sold high — creates a rebalancing premium
    - Fix: tilt each donor's weight on day t by its gross return over the prior k days
    - Extended version of ABK (2013)
- Intercept / unit fixed effect (gsynth, SDID, demeaned SC)
  - Differences out every unit-level average
  - Carries a pre-period mean forward (Ashenfelter dip)
- Coarser frequency
  - Monthly data bills the noise 12x/yr, not 250x
  - Leaves: attenuates $\approx$ 10x, does not immunize

Matters most when few treated stocks per event period

M&A placebos: correcting the donor leg removes the drift



Additive CATT at +250 days, 14,618 deals, 18-month block bootstrap:

	SC	SC + ABK(5)
Placebo	−30.2	+1.8 [−3.4, 6.4]
Treated	−32.1	−3.8 [−9.2, 0.9]

- Weights, pre-fit, and estimand unchanged — only evaluation changes
- Announcement effect *strengthens* to +1.2% and now persists through +21 days (uncorrected SC showed −1.5%: noise drag swamped it)
- Corrected gap agrees with gsynth runup-corrected estimate to 0.6pp

Calendar-time portfolio: formal decomposition

- Calendar-time α decomposes as:

$$\alpha^{\text{cal}} = \sum_{s \in \mathcal{S}} \sum_{\kappa=0}^{H} \omega_{s,\kappa}^{\text{cal}} \cdot \tau^{\text{ATT}}(s, s+\kappa) + |\alpha|^{\text{cal}} + \text{bias}^{\text{cal}}$$

where $\omega_{s,\kappa}^{\text{cal}} = \frac{n_s}{T_{\text{cal}} \cdot N_{s+\kappa}}$

- When treatment changes factor loadings:
  - Event-time: τ^{ATT} includes $\left(\beta_i^{\text{post}} - \beta_i^{\text{pre}}\right) F_t$
  - Calendar-time: estimates anomaly relative to *new* risk profile